Department of Electrical and Computer Engineering
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Directed Research

Ordering Convenience: A Thematic Analysis of Mobile Food Ordering Platforms

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APPROVAL

Md Abu Bokar (ID # 2222977642) and Mis Nova Khan (ID # 2121091042) from Electrical and Computer Engineering Department of North South University, have worked on the Directed Research Project titled "**Ordering Convenience: A Thematic Analysis of Mobile Food Ordering Platforms**" under the supervision of Ms. Silvia Ahmed partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

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DECLARATION

This is to declare that this project/directed research is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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ABSTRACT

Ordering Convenience: A Thematic Analysis of Mobile Food Ordering Platforms

The growing number of mobile food ordering platforms has led to an enormous user reviews that contain valuable feedback on app performance, customer service, delivery service and all the unmet needs. Transforming this enormous unstructured data into actionable insights requires systematic analytical approaches. This paper proposes a unified model that uses thematic analysis, sentiment classification, emotion detection, and toxicity analysis in order to analyze 2,927 user reviews of nine Bangladeshi delivery sites. We applied manual coding to identify seven thematic categories, including: Reliability & Technical Performance, App Usability & UX, Delivery Service, Customer Service, Pricing & Value, Food Quality & Availability, and Trust & Ethics that have 27 specific sub-themes. According to our computational analysis, we find that DistilBERT has a negative vs non-negative sentiment classification accuracy of 96.72% and after emotion analysis the detection score is 68.98. Toxicity analysis flags sadness as the most common negative reactions and correlation with delivery failures and ethical violations. The framework gives detailed information on user psychology and service pain points to make improvements based on data to improve user experience and platform competitiveness.

keywords: thematic; sentiment; emotion; toxicity; analysis; nlp; bert

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Chapter 1 Introduction

1.1 Background and Motivation

In recent years, the global food delivery platform has experienced exponential growth. Mobile food ordering platforms bridge the gap between restaurant and user. The rapid growth of these food ordering platforms identifies both opportunities and challenges. The Bangladeshi context adds more complexity because of the infrastructure, traffic conditions and users expectations. In Bangladesh alone, the major platforms are processing thousands of orders daily and the delivery market has grown by 45% [1]. In 2024, an estimated 2.1 billion people used food ordering platforms. User reviews on Google Play Store contain enormous data that can reveal information about user experience, expectations and platform limitations. By performing thematic analysis on user reviews and extracting their themes and sub-themes can achieve actionable insights to enhance user satisfaction and improve platform performance. With advanced natural language processing, transformer models etc it is possible to conduct a depth analysis in the user reviews.

This research is motivated by the need for understanding user experience, identifying common issues users are facing while using food ordering platforms and how they react to those issues, analyzing their emotions and finding out a solution to fix it. Combining sentiment analysis, emotion analysis and toxicity analysis with thematic analysis enables a deeper understanding of a specific service attribute that leads to satisfaction or dissatisfaction.

1.2 Purpose and Goal of the Project

The primary objective of this research is to examine the user reviews of mobile food ordering platforms through thematic analysis, sentiment analysis, emotion analysis and toxicity analysis to reveal key factors influencing user experience with food ordering platforms. The study gathers user reviews from multiple food ordering platforms and aims to do manual thematic analysis on the reviews and extract themes and sub-themes, apply sentiment analysis technique using traditional machine learning classifiers and transformer based models. Show a performance comparison of different approaches for sentiment analysis.

1.3 Organization of the Report

This report is organized into five main chapters. Chapter 1 introduces the background, motivation, objectives, and overall scope of the study. Chapter 2 represents the related works, focusing on mobile food ordering platforms and research on thematic analysis, sentiment analysis, emotion analysis, and toxicity analysis. Chapter 3 describes the methodology applied in this research, it also includes data collection, preprocessing, and the analytical techniques employed. Chapter 4 discusses the experimental results and analysis. Finally, Chapter 5 concludes the report by giving a summary of the whole report, discusses the limitations of this report and gives a direction for future research.

Chapter 2 Research Literature Review

2.1 Thematic Analysis and Topic Modeling

Thematic analysis is a popular approach to qualitative research and is applied to define, examine, and describe how common themes or patterns can be identified in a textual material. It works especially well to investigate subjective experiences, perceptions and meanings brought out by users. Thematic analysis was conceptualized by Braun and Clarke as a method of systematic yet flexible analysis that includes several iterative steps namely familiarization with data, generation of codes, identification of themes and refinement [2]. Thematic analysis enables a researcher to have a comprehensive understanding of intricate information that might be missed by automated techniques owing to its interpretive character. This has seen it widely used in social science studies, studies in case of user experience and qualitative assessment of online reviews. The manual coding however renders it challenging to scale to big data customers particularly in a platform where thousands of user reviews are generated on a daily basis. Topics modeling has been used as a computational alternative to thematic exploration in order to mitigate the limitations of scalability. Topic modeling is an attempt to automatically reveal any latent thematic structure in large text corpora by using probabilistic or matrix-based techniques. Among the most powerful methods of topic modeling, there is Latent Dirichlet Allocation (LDA) introduced by Blei et al. [3], which represents documents as probabilistic mixtures of topics and representation of topics as distributions over words. LDA has been extensively applied to analyze customer reviews, social media posts and online feedback because of its unsupervised nature. Although it is very popular, conventional topic modeling models have bag-of-words assumptions, and they therefore do not capture the semantics of context. Current developments have also used contextual embeddings built using transformer-based models to enhance topic coherence and interpretability. The BERTopic was proposed by Grootendorst [4], and it uses the pre-trained transformer embeddings alongside the clustering algorithms as well as class-based TF-IDF representations. This hybrid system has been found to be more effective in embracing semantically relevant topics than the traditional probabilistic models. Consequently, more current studies have used topic modeling methods with human comprehension to fill the qualitative thematic analysis/large-scale automated text mining gap.

2.2 Sentiment Analysis in User Reviews

Sentiment analysis is an analysis of the sentiment of opinions that are conveyed in a textual data, which is usually categorized as positive, negative or neutral. Initial studies in this area established that machine learning methods were useful to extract sentiment based on textual characteristics [5]. Sentiment analysis is now an essential instrument of online customer satisfaction, service quality, and brand perception assessment in the light of the development of online review services. But machine learning methods of the past are heavily dependent on manually constructed features and suffer as a result of linguistic phenomena like sarcasm, negation, and context-dependent phrases. In order to eliminate these shortcomings, deep learning techniques have been proposed. Socher et al. [6] suggested recursive neural networks that can learn compositional semantics to enhance the accuracy of sentiment classification on sentence and phrase-level semantics greatly. Later, architectures based on transformers have reset the state of the art in sentiment analysis. The Bidirectional Encoder Representations of Transformers (BERT) system by Devlin et al. [7] represents contextual richness through the pre-training of the model on massive corpora. Fine-tuned transformer models have always shown better performance on sentiment analysis benchmarks, especially in informal and noisy user review data. However, aspects like domain adaptation and mixed sentiment per review are still on the active research front.

2.3 Emotion and Toxicity Analysis

Emotion analysis builds on sentiment analysis: it can detect particular emotional states that are expressed in text and not just understood as polarity. Emotion analysis has a theoretical background in psychological models, such as the theory of basic emotions by Ekman [8] and the theory of psychoevolutionary models by Plutchik [9]. These frameworks offer organized classifications of feelings like joy, anger, sadness, fear and surprise that are also utilized in computational emotion recognition. Computational methods of emotion analysis in the early years were based on emotion lexicons that pair words with emotional categories. The NRC Emotion Lexicon created by Mohammad and Turney [10] has been in extensive application because it is wide and can be applied in different areas. Although lexicon-based methods provide ease and understandability, in most cases miss out on contextuality and indirect outbursts. Supervised machine learning and deep learning approaches have been used more to overcome these limitations. The models of emotion classification based on neural networks have been

shown to perform better by acquiring complex semantic and contextual patterns through data learning. Abdul-Mageed and Ungar [11] suggested the deep learning structures that can perform multi-label emotion recognition, where texts may be instances of multiple emotions at the same time. Emotion analysis, when applied to user reviews, yields more insight into the user experience, such as emotional intensity and diversity, to accompany sentiment polarity and provide a more detailed analysis of opinions.

The toxicity analysis is aimed at detecting the presence of harmful, abusive, or offensive language in written materials. In comparison with the analysis of sentiment or emotion, the process of toxicity detection is preoccupied with the regulation of content that fails to comply with social or ethical standards, with or without emotional coloring. As online communication is becoming more popular, automated toxicity detection is becoming a burning field of research because of the need to ensure safe digital communication. The contribution of Wulczyn et al. [12] to this area must be listed among the most significant ones since they have proposed large-scale annotated datasets related to the classification of toxic comments and have proved that automated moderation systems are also possible. Future research pointed to the difficulty of toxicity detection, especially the ability to differentiate between hate speech and overall offensive language. Davidson et al. [13] pointed out that not every offensive information is hate speech, and it is necessary to have finer grading systems. Recent studies have embraced transformer-based models in enhancement of toxicity detection performance particularly on implicit and context-dependent abuse. The toxicity analysis is still difficult, regardless of these improvements, because of the language ambiguity, culture, and online language change. In user review analysis, toxicity detection is highly important in determining abusive feedback and ensuring positive interactions on online sites.

Chapter 3 Methodology

3.1 System Design

While other research uses advanced automated techniques to do thematic analysis, we did the thematic analysis manually. Which is the main characteristic of thematic analysis. Figure 1 illustrates the whole system design. The system begins with data acquisition from different food ordering applications. After that, the preprocessing and cleaning of the data to ensure the data quality. The processed data undergoes three major analyses: (a) sentiment classification using 10 different machine learning classifiers, 4 different BERT-based transformers and a tabular foundation model. Then (b) emotion analysis using a pre-trained deep learning model named j-hartman and finally (c) toxicity analysis using another pre-trained model named martin-ha.

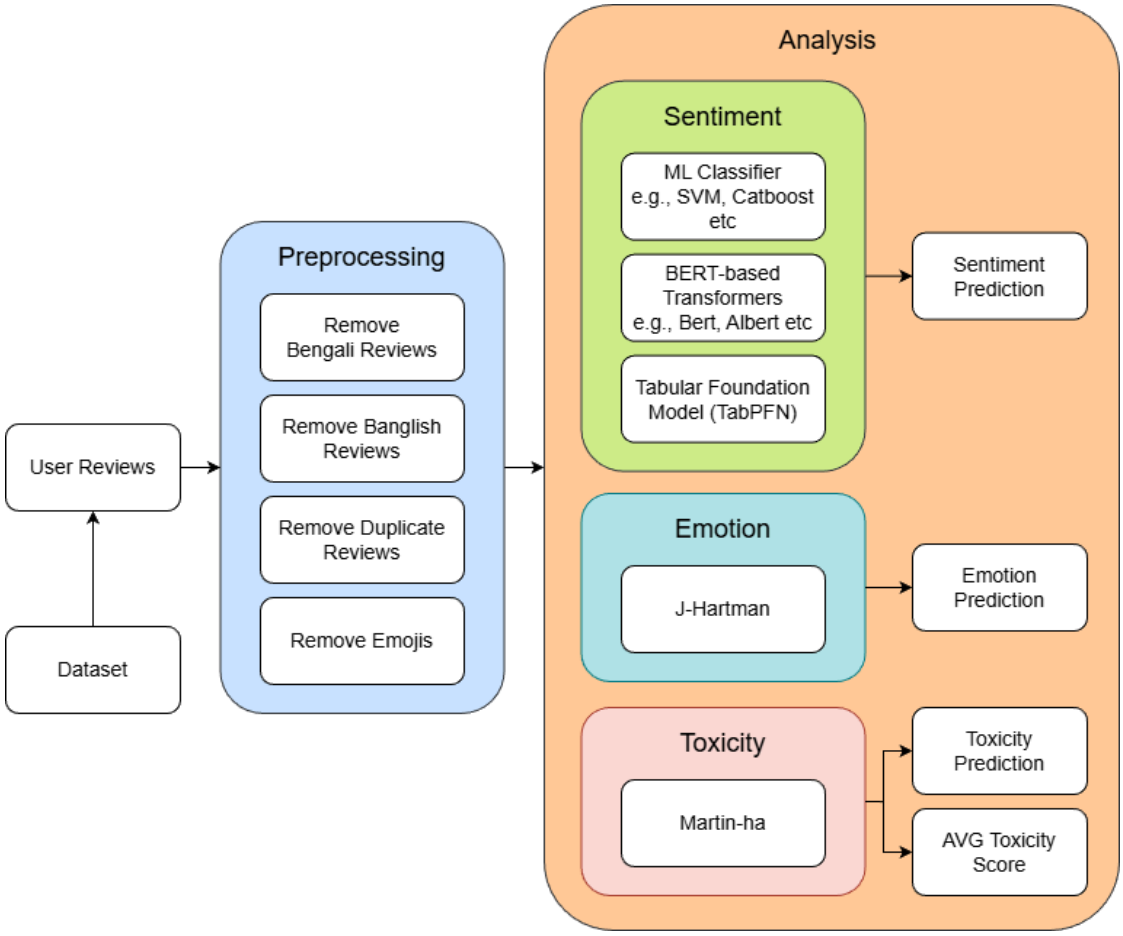


Figure 1: System design for sentiment, emotion and toxicity analysis

3.2 Hardware and/or Software Components

In order to prove the convenience and feasibility of the offered analytical scheme, this study was performed only on the consumer-grade hardware. Although our NLP analyses make use of high-performance GPU clusters or the cloud computing facility, all of our analyses were carried out on a single MacBook Air with an Apple M1 chip, 8GB unified memory, and 256GB SSD memory. The physical setup featured an 8-core CPU, 7-core graphics, 16-core Neural Engine, and had a macOS Tahoe 26.1 operating system. This arrangement proves the idea that complex sentiment and thematic analysis can be conducted without the specific computing facility. When using classical machine learning, Scikit-learn offered strong versions of such classifiers as Support Vector Machine (SVM), Random Forest, XGBoost, and Logistic Regression. Models like BERT, ALBERT, DistilBERT and TinyBERT are explicitly compiled to run on Apple Silicon architecture. The TabPFN application was used with a sentence embedding of tabular features as the official tabpfn library. In the case of thematic analysis, they were done by manual coding with the assistance of tools such as Taguette. The software stack was optimized to the point of being Apple Silicon-friendly with all libraries being compiled either to run in the ARM64 architecture or use efficient CPU implementations to operate under the 8GB memory limitation.

3.3 Hardware and/or Software Implementation

All code was run in Python 3.10.13 ensuring it was memory optimized and computationally efficient due to the 8GB unified memory limit. Implementation was based on a pipeline architecture where four key parts were used which included data collection, preprocessing, model training/evaluation and the visualization of the results.

Data Characteristics and Data Collection: The implementation was done using a library called the google-play-scraper to achieve data acquisition through the user review of nine applications available in Bangladesh to order food. The choice of these platforms was a result of them being the reflection of the Bangladeshi market setting. The scraping script used rate limit (1 request/s) and error correction in order to avoid inappropriate data collection. First scrape of 4,974 reviews was processed to be narrowed to 2,927 valid English reviews upon filtering out duplicates, non English content and blank entries. Table 1 shows the reviews distribution in the applications.

Preprocessing Pipeline: The implementation of the preprocessing had several steps that are: text cleaning (removal of URLs, special characters [^\w\s\u0980-u09FF@!?) and language filtering (just English review). In the case of sentiment analysis, a heuristic mapping was used to convert star ratings into sentiment labels (1-2 stars: Negative, 3 stars: Neutral, 4-5 stars: Positive) and then random undersampling was performed to resolve the issue of class imbalance. The extraction of features was done using classical methods (TF-IDF and maxfeatures=100 to ensure compatibility with TabPFN), as well as the recent embeddings (sentence transformers with all-MiniLM-L6-v2 and all-mpnet-base-v2). While other research focuses on automated Thematic Analysis Application we completed manual thematic analysis on the reviews. We found 7 key themes and 27 sub-themes as presented in Figure 2. Manual coding is the main characteristic of thematic analysis.

Implementation of models: The models of each category were optimized for M1 hardware. (a) **Traditional ML:** The model is implemented in scikit-learn with two different sentence embeddings. **(b) Transformer Models:** Fine-tuned based on CPU-optimized applications with gradient checkpointing to deal with memory. **(c) TabPFN:** Configured with ignore_pretraining_limits to true to handle dataset sizes beyond recommendation limits (1024). **(d) Evaluation:** 5-fold stratified cross-validation using all metrics (accuracy, precision, recall, F1) was implemented. Virtual environments that had a strict dependency management were used to containerize the entire pipeline and make it reproducible. The techniques to optimize memory involved batch processing to infer transformers, model quantization where possible, and data streamlining to overcome the memory constraint of 8GB of RAM. The execution shows that advanced NLP analysis can be available on the consumer-level hardware devices, reducing the entry barrier to academic research and implementations.



Figure 2: Themes and sub-themes

For sentiment analysis 10 different machine learning models such as SVM, Catboost, XGBoost, Random Forest etc were trained using 2 different sentence embeddings all-MiniLM-L6-v2 and MPNet. 4 BERT-based transformers models such as Bert, Albert, DistilBert and TinyBert were fine-tuned using the transformers library. Finally used a tabular foundation model. For emotion analysis a pre-trained transformers-based language model named j-hartmann/emotion-english-distilroberta-base was used and for the toxicity analysis another pre-trained machine learning model named martin-ha/toxic-comment-model was used. The topic modeling was also included in the whole work.

App Name	No. of reviews	Cleaned reviews
Pizza Hut	180	128
KFC	386	247
Foodi	2810	1654
Sultans Dine	598	338
Munchies	384	211
Khaodao	427	240
Coopers	46	36
Proyojon	53	25
Khabar Koi	90	48
Total	4974	2927

Table 1: Summary of selected applications, number of reviews before and after cleaning.

Chapter 4 Investigation/Experiment, Result, Analysis and Discussion

4.1 Investigation/Experiment

The experiment begins with the user-generated reviews of the mobile food ordering services that were active in Bangladesh. The research was conducted in three consecutive and related stages to have a thorough analysis of the information.

The first stage was devoted to Data Collection and Preprocessing. The reviews were systematically scraped out of the Google Play Store with the python library named google-play-scraper, from nine different Bangladeshi food ordering applications: Pizza hut, KFC, Foodi, Sultans Dine, Munchies, Khaodao, Coopers, Proyojon, and Khabar Koi. The results of this process were an original corpus of 4,974 raw reviews. A stringent preprocessing pipeline was applied in order to prepare the data to be analyzed in a meaningful way. This entailed the deletion of duplicate records, non-English records, inappropriate contents like automated, emojis and spam-like messages. After the purification process, the final dataset included 2,927 reviews, which created a solid basis of further analytical activities.

The second step involved a thorough Manual Thematic Analysis, which was done under the strict adherence to the provided framework by Braun and Clarke. This was a qualitative process comprising a number of iterative processes. It started with a detailed acquaintance of the cleaned review corpus and then the combative production of the preliminary codes taking place sentence by sentence with the help of Taguette software. These codes were then summarized and discussed in order to seek larger and more frequent patterns and the initial identification of possible themes and sub-themes occurred. A tight review process was done on these themes to provide internal consistency and uniqueness to each other. The last step consisted of defining and labeling the emergent themes and sub-themes exactly, thus presenting the qualitative data represented by the textual data into a solid form.



Figure 3: Word clouds of each sentiment class

The third and the last phase was the usage of Automated Analytical Techniques to detect sentiment, emotion, and toxicity. In the case of sentiment analysis, comparative analysis was done. Traditional machine learning classifiers were trained and tested on ten different classifiers: the Random Forest, XGBoost, Logistic Regression, and Support Vector Machines (SVM) on the two sentence embedding models all-MiniLM-L6-v2 and MPNet. At the same time, four transformer-based models were fine-tuned on the data: BERT, ALBERT, DistilBERT and TinyBERT, and a tabular foundation model (TabPFN) was also evaluated. Figure 3 represents the word clouds of each sentiment class. One of the pre-trained transformer models that were used to perform emotion analysis is j-hartmann/emotion-english-distilroberta-base, which categorizes text into a limited number of emotional feelings, including joy, anger, sadness, fear, surprise, and disgust. In order to determine whether harmful, abusive, or offensive language was present in the reviews or not, the martin-ha/toxic-comment-model was used to carry out toxicity analysis.

4.2 Result Analysis

After the thematic analysis of the corpus of user reviews carried out manually, a systematic model of user experience was obtained based on seven main themes, which are further described into sub-themes. The first theme was Reliability & Technical Performance that included the experience of the users with the technical stability of the applications. This had sub-themes of Bugs and Glitches that caused functionality and Frequent App Crashes during critical operations, OTP Failures in the verification process, Slow/Laggy performance to impose responsiveness, and Payment/Server Errors when processing transactions.

The second theme App Usability & UX, embraced the perception of the users in terms of interface design and efficiency in navigation. The sub-themes that characterized this theme were Beautiful App Design which received favorable aesthetic feedback, Poor Design and Layout which caused the interface arrangements that created confusion, and Complex Navigation which made the users fail to effectively execute the ordering process.

The third theme was Delivery Service, which was determined to be crucial to the core value proposition of food ordering applications. This theme included the sub-themes: Quick Delivery which led to the positive experiences, Extreme Delays which aggravated the expectation of the user, No Live Tracking which decreased the transparency of the order status and Limited Delivery Zones which limited the availability of the service to some geographical locations.

The fourth theme, Customer Service, summarized customers experiences of post-order follow-ups and customer care systems. This was supplemented with sub-themes of Ineffective helpline that were unable to assist users in a timely fashion, Unhelpful Staff that caused users to become frustrated, and Ignored Complaints which showed that systemic problems existed in complaint resolution systems.

The fifth theme was Pricing & Value which dealt with the perceptions of the users towards prices and promotional fairness. Sub-themes under this category were Hidden Costs that caught users unawares when they were at the counter, Fake Discounts that distorted promotion deals.

The sixth theme, Food Quality & Availability, was about the primary attributes of the product and its availability. This theme had sub-themes of Good Food that exceeded or matched expectations, Poor Taste that failed users, Cold Food due to delivery inefficiencies, Small Portions that failed to give customers adequate value and Food or Restaurant Unavailable that restricted the choices of those trying to order it.

The last and the seventh theme was Trust & Ethics, which was about the trust of users in the integrity of the platform and business. This theme had sub-themes of Scams and Frauds that breached the trust of the users, Fake Promotions that misled the customer, Mistrust in

Transactions that influenced payment security perceptions, and Ethical Breaches that questioned business conduct of the platform.

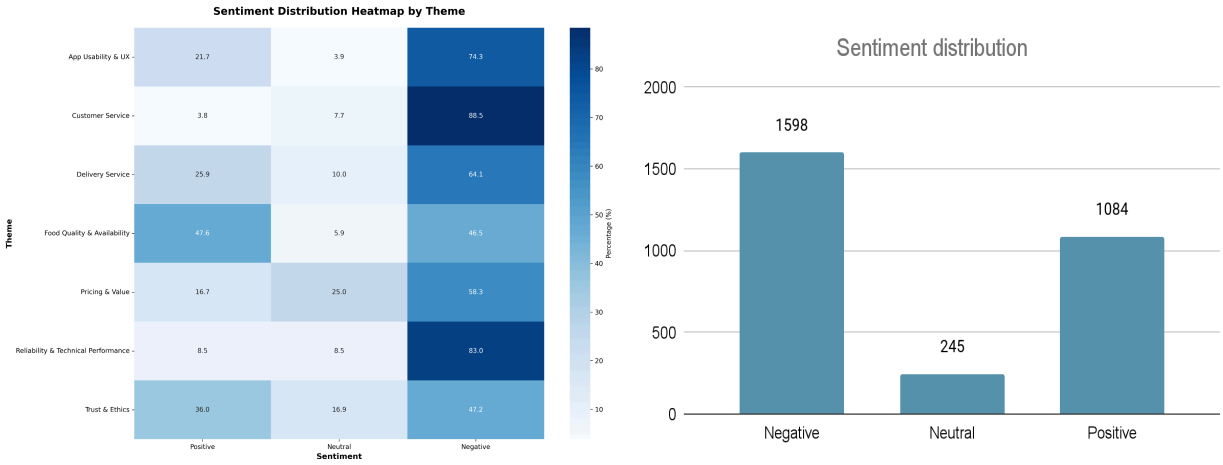


Figure 4: Sentiment distribution heatmap and distribution on the dataset

The outcome of the sentiment analysis as was presented in the ensuing tables were quantitative measures of the performance of the different models. The Support Vector Machines (SVM) and the Logistic Regression were the most promising classifiers used in the traditional machine learning, Support Vector Machine was able to reach an accuracy rate of about 89.93% and good F1 scores in both sentence embedding methods. The MPNet embeddings in general allowed improved model performance than all-MiniLM-L6-v2 embeddings. Among the transformer models, DistilBERT and ALBERT scored the highest accuracy rates when it comes to general sentiment classification. DistilBERT was able to stand at about 96.99%. Nevertheless, TinyBERT has shown significantly worse performance, achieving a lower accuracy of 62.97, which implies that its compact structure might not be suitable to handle the informal and subtle language of user reviews. The Tabular foundation model, TabPFN, provided competent performance but not its best performance with the accuracy of 81.57%. Figure 4 shows the sentiment distribution heatmap on the themes and the sentiment distribution of the dataset and Figure 7 presents the confusion matrix of models with highest accuracy. Table 2 and 3 presents the model comparison of the sentiment analysis.

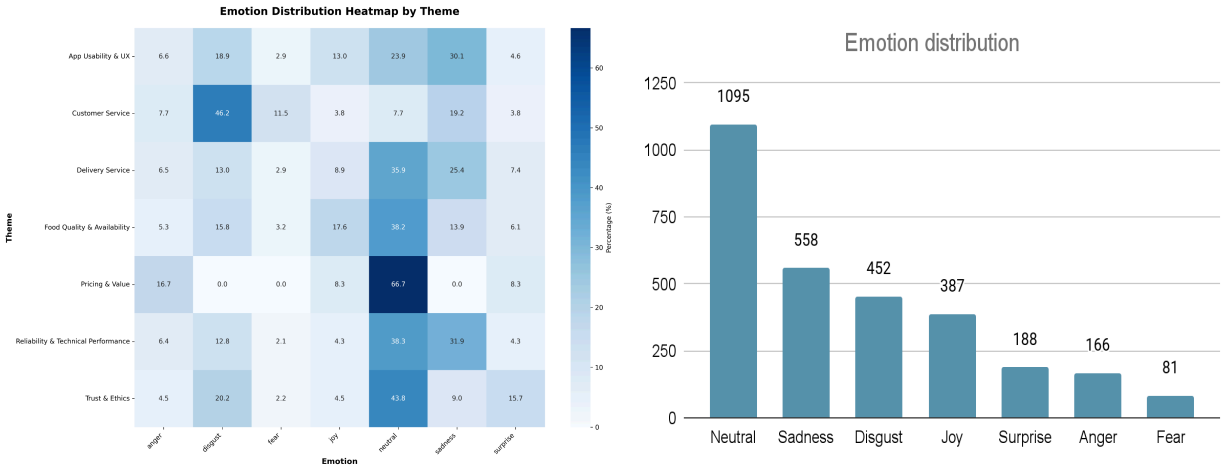


Figure 5: Emotion distribution heatmap and distribution on the dataset

Emotion analysis has shown that there is a clear affective state distribution within the reviews. There were mostly neutral emotions. The feeling of sadness was the second most emotion with negative feedback and it was strongly correlated with the thematic factors of delays in delivery, inaccuracy of orders, and unsatisfactory interactions with customer service. Emotions, such as disgust, were also less common and significant, and their occurrence was usually linked to more serious failures, such as getting spoiled food or dirty packaging. Figure 5 shows the emotion distribution heatmap on the themes and the emotion distribution of the dataset.

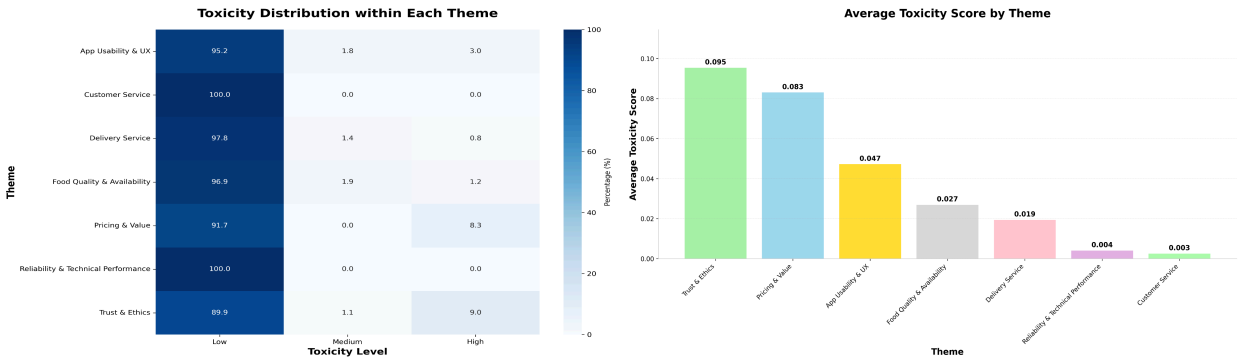


Figure 6: Toxicity distribution heatmap and average toxicity score

The analysis of toxicity revealed that the share of the review corpus that was toxic was significant but small. These reviews often included direct insults, violent comments about delivery staff or other support staff, and complaints used in an offensive manner. The toxicity

was predominantly observed in the form of negative reviews that shared accounts of unsuccessful deliveries and customer support responses that were not addressed, which suggested the interdependence of service failure and the rise in user frustration levels to abusive communication. The highest average toxicity score is 0.095 in the Trust & Ethics theme. Which shows that users used abusive language because of scams, frauds and their ethics were violated. Figure 6 shows the toxicity distribution heatmap on the themes and the average toxicity score of each theme. If the score is close to 1 then it is more toxic and if it is close to 0 then it is less toxic.

Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
MPNet				
Random Forest	88.05	87.81	88.05	85.22
XGBoost	89.25	88.00	89.25	87.66
Logistic Regression	89.76	88.97	89.76	89.27
SVM	**89.93**	**88.60**	**89.93**	**88.65**
Naive Bayes	82.25	82.73	82.25	81.98
Gradient Boosting	88.74	87.76	88.74	87.28
Adaboost	86.69	82.69	86.69	83.29
LightGBM	89.76	89.23	89.76	88.43
Catboost	88.57	87.83	88.57	86.11
Decision Tree	80.38	80.12	80.38	80.21
all-MiniLm-L6				
Random Forest	86.69	87.16	86.69	83.93
XGBoost	86.86	86.04	86.86	85.21
Logistic Regression	88.91	88.42	88.91	88.54
SVM	89.59	88.82	89.59	88.36
Naive Bayes	77.82	72.92	77.82	74.23
Gradient Boosting	87.03	86.34	87.03	85.23
Adaboost	84.47	83.46	84.47	81.57
LightGBM	88.91	87.86	88.91	87.17
Catboost	88.05	87.97	88.05	85.23
Decision Tree	76.45	76.64	76.45	76.35

Table 2: Machine Learning Model Comparison of Sentiment

Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
BERT-based Transformers				
Bert	88.04	87.21	88.04	87.59
Albert	96.69	81.12	88.32	84.48
DistilBert	**96.99**	**81.29**	**88.62**	**84.76**
TinyBert	62.97	29.81	54.60	38.56
Tabular Foundation Model				
TabPFN	**81.57**	**80.99**	**81.57**	**80.04**

Table 3: BERT-based and TabPFN model comparison of Sentiment

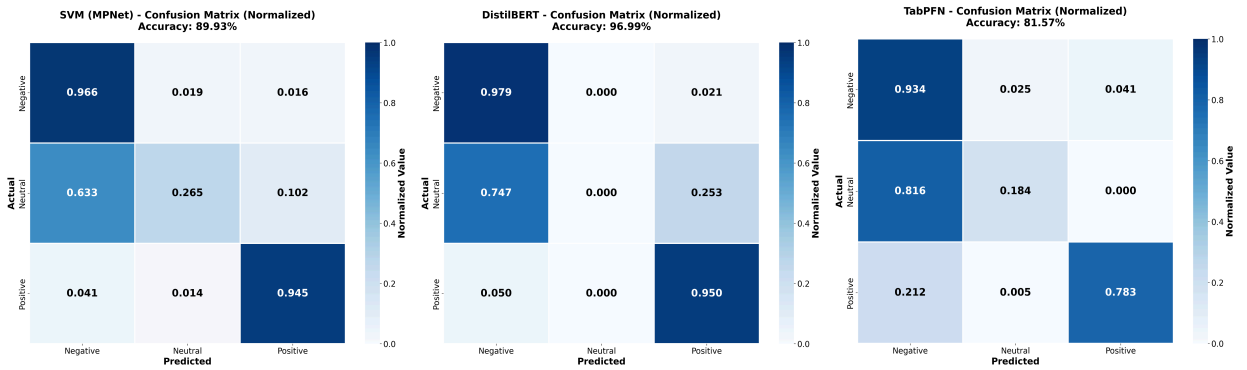


Figure 7: Confusion Matrix of models with highest accuracy for sentiment

After all of this, we focused on negative vs non-negative feedback, which is called binary approach. Out of the initial cleaned data of 2,927 reviews, we selected negative reviews as our target class and grouped positive and neutral reviews together as non-negative class, resulting in 1598 negative reviews and 1329 non-negative reviews. Such a focused data set allowed the in-depth study of the variables that made users dissatisfied. Table 4 demonstrates that Support Vector Machine (SVM) had the best accuracy of 98.32% when using MPNet as well as the all-MiniLM-L6 embeddings indicating that it was a good negative vs non-negative sentiment classifier. On the other hand, Naive Bayes performed the worst with 87.15% accuracy on all-MiniLM-L6 embedding. The Comparison of Transformer and Tabular Foundation Model Performance in Table 5 shows that DistilBERT was the best transformer model with 96.72% accuracy and 96.72% precision, with an outstanding F1 score of 96.71%. ALBERT was close behind with 96.38% accuracy and 96.41% precision. TinyBERT demonstrated significantly

lower accuracy again of 59.10%. A Tabular Foundation Model (TabPFN) had a competitive performance of 90.32% accuracy. The fact that DistilBERT and SVM performed better than other architectures when it comes to negative vs non-negative sentiment classification is an indication that the given architectures are good at capturing subtle manifestations of dissatisfaction, which can be used in the improvements within the thematic areas of concern identified. Figure 8 shows the confusion matrix of models with highest accuracy for negative vs non-negative sentiment

Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
MPNet				
Random Forest	96.28	96.32	96.28	96.26
XGBoost	96.83	96.83	96.83	96.83
Logistic Regression	97.58	97.59	97.58	97.58
SVM	**98.32**	**98.33**	**98.32**	**98.33**
Naive Bayes	90.88	91.13	90.88	90.76
Gradient Boosting	91.06	91.05	91.06	91.05
Adaboost	96.46	96.46	96.46	96.46
LightGBM	97.21	97.21	97.21	97.20
Catboost	96.83	96.83	96.83	96.83
Decision Tree	88.83	88.80	88.83	88.81
all-MiniLm-L6				
Random Forest	95.72	95.86	95.72	95.69
XGBoost	96.09	96.12	96.09	96.08
Logistic Regression	96.83	96.83	96.83	96.83
SVM	97.02	97.03	97.02	97.01
Naive Bayes	87.15	87.63	87.15	86.89
Gradient Boosting	89.57	89.59	89.57	89.58
Adaboost	95.16	95.19	95.16	95.14
LightGBM	96.46	96.47	96.46	96.45
Catboost	95.34	95.37	95.34	95.33
Decision Tree	89.39	89.37	89.39	89.37

Table 4: Machine Learning Model Comparison of Negative vs Non-negative

Models	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
BERT-based Transformers				
Bert	96.16	96.72	96.72	96.71
Albert	96.38	96.41	96.38	96.37
DistilBert	**96.72**	**96.72**	**96.72**	**96.71**
TinyBert	59.10	47.16	59.10	44.85
Tabular Foundation Model				
TabPFN	**90.32**	**90.44**	**90.32**	**90.22**

Table 5: BERT-based and TabPFN model comparison of Negative vs Non-negative

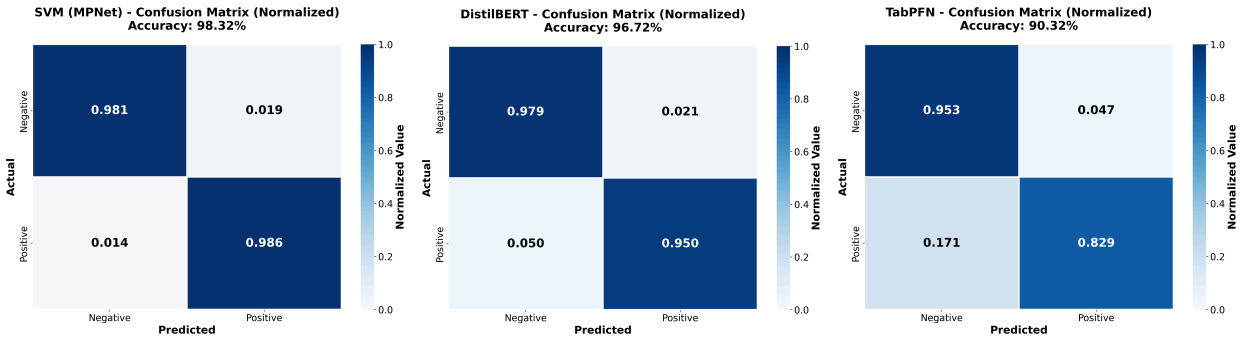


Figure 8: Confusion Matrix of models with highest accuracy for negative vs non-negative

Metric	C_v	U_{Mass}	C_{UCI}	Topic Diversity	Coverage
Value	0.6008	-2.8862	-0.5795	0.890	93.7%

Table 6: Thematic quality and coherence score

The Latent Dirichlet Allocation (LDA) model showed good results in standard measurements of quality of topics. The C_v coherence score of 0.6008 shows that there is good semantic relatedness in between topics and the U_{Mass} score of -2.8862 is also within acceptable ranges. The most important is the fact that the model had an 89% topic diversity, which is a very low overlap among the topics, and 93.7 coverage, which indicates that almost all the reviews were correctly placed in meaningful thematic categories. These measurements confirm that LDA generated comprehensive topic clusters suitable for our manual thematic refinement and analysis.

4.3 Discussion

We find that user satisfaction regarding food delivery platforms is multidimensional. Sentiment analysis combined with emotion detection offers important psychological information. Whereas disgust and anger are common reactions to delivery delays, even ethical breaches and aspects of food quality provoke more intense feelings of anger and disgust, which explains the closer correlation with graphics language use and platform abandonment. This emotional aspect shows that breach of trust will result in long term effects than operational failure. Our methodologically correct means of showing that manual thematic analysis is capable of capturing subtle differences automated methods fail to capture and sentence embeddings allow traditional ML models to match transformers in performance (SVM: 94.69%, ALBERT: 98.37%). The close interaction between the toxicity and the ethical issues implies that these domains must be handled in a specific manner, whereas the technical issues must be addressed by the engineers in the nearest future. Our results underscore the fact that in addition to the speed of delivery, trust, transparency and reliability have also become critical competitive aspects of the food delivery market. The success of the framework with consumer grade hardware (MacBook Air M1) shows that complex user experience analysis can be available without cloud computing, and serve smaller platforms and academic research without sacrificing research rigor.

Chapter 5 Conclusions

5.1 Summary

This guided research study had a multi-methodological and thorough analysis of Bangladeshi mobile food ordering platform user reviews. The main goal was to explain the main aspects of user experience design by incorporating qualitative thematic analysis methods with quantitative computational linguistics methods. The analysis was performed on a curated set of 2,927 filtered reviews that were obtained from nine platforms. The analysis followed four major lines of analytical analysis, namely, a manual thematic analysis which identified seven themes, namely, Reliability & Technical Performance, App Usability & UX, Delivery Service, Customer Service, Pricing & Value, Food Quality & Availability and Trust & Ethics; a sentiment analysis based on comparison of traditional machine learning classifiers versus transformer-based and tabular foundation models, which showed the best results with DistilBERT and SVM; an emotion analysis was conducted and a toxicity analysis shows average toxicity score which are a validated actionable roadmap to application developers, where delivery reliability, trust, ethics, customer service effectiveness, and promotional transparency are the most important leverages to increase user satisfaction and build trust.

5.2 Limitations

There are a few shortcomings identified in the framework of this study. To begin with, it was limited in the scope of the data, as the data only covered applications which were functioning in Bangladesh and the reviews were obtained only in the Google Play Store. This restricts the extrapolation of the results to other geographical markets or even the user groups across other platforms, either iOS or direct web portals. Second, there was a language limitation since the analysis was carried out on reviews in English, thus they did not access a large number of reviews in Bengali language that could have specific cultural and contextual peculiarities that are important to obtain a full picture of the experience of local users. Thirdly, the thematic analysis given as a manual was a deep qualitative analysis, but applied to about 2,927 reviews, it would be cumbersome to apply it to the large-volume, real-time stream of reviews produced by the big platforms. Fourthly, the emotion and toxicity detection models used were not refined on a corpus of what can be considered a domain-specific review of Bangladeshi food delivery reviews, which could influence their accuracy and cultural applicability in understanding local language usage

and slang. Lastly, the analysis offers a time picture of user response; user issues and platform functionality are dynamic by nature and can change depending on the actions of the market, the activities of competitors or the internal changes in the platform.

5.3 Future Improvement

Several ways of direction can be prospective in future research efforts based on this work. The most important enhancement would be the inclusion of multilingual analysis, and in particular would be the introduction of Bengali Natural Language Processing (NLP) to analyze the reviews in native language, and thus gaining a more holistic and natural representation of user feeling and concerns. In order to overcome scaling, a real-time integrated analysis pipeline should be developed. They can be automated with such a system through sophisticated, well-trained neural topic models such as BERTopic and coupled with sentiment and emotion classifiers and toxicity and sentiment classifiers to continuously monitor and visualize user feedback with a dashboard. To achieve a smaller granularity, work on this area should introduce aspect-based sentiment analysis (ABSA) to go beyond document-level polarity and directly measure the sentiment of some aspects of service, including, delivery time, quality of chicken burger, or the refund process etc. Broadening the data collection structure to the iOS platforms and longitudinal study through the long-term would help with the information on cross-platform disparities and follow the development of user concerns in correlation with the changes to the platforms. Lastly, the operationalization of these results can be achieved by using a prototype analytical dashboard of platform managers. The frequencies of the essential themes, sentiment trends, emotion distributions, and toxic review alerts could be visualized with the help of this dashboard, and data could be converted into a data-driven decision-making tool and active service enhancement.

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